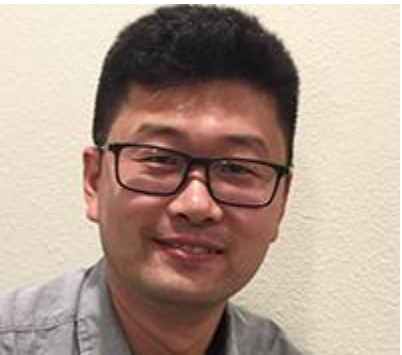


# Competitive Coding

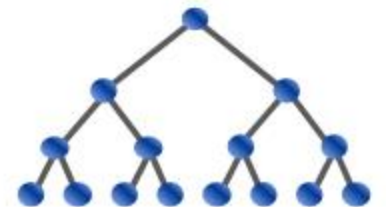
cs4fun, Paul Cao

# Welcome

- Goal: prepare our talents to be ready for competitive coding.
- Approach: solid foundation, fun experience, rigorous curriculum
- Expectation: participate during class, complete homework assignments each week
- Questions?



**USA Computing Olympiad**



# Introduction

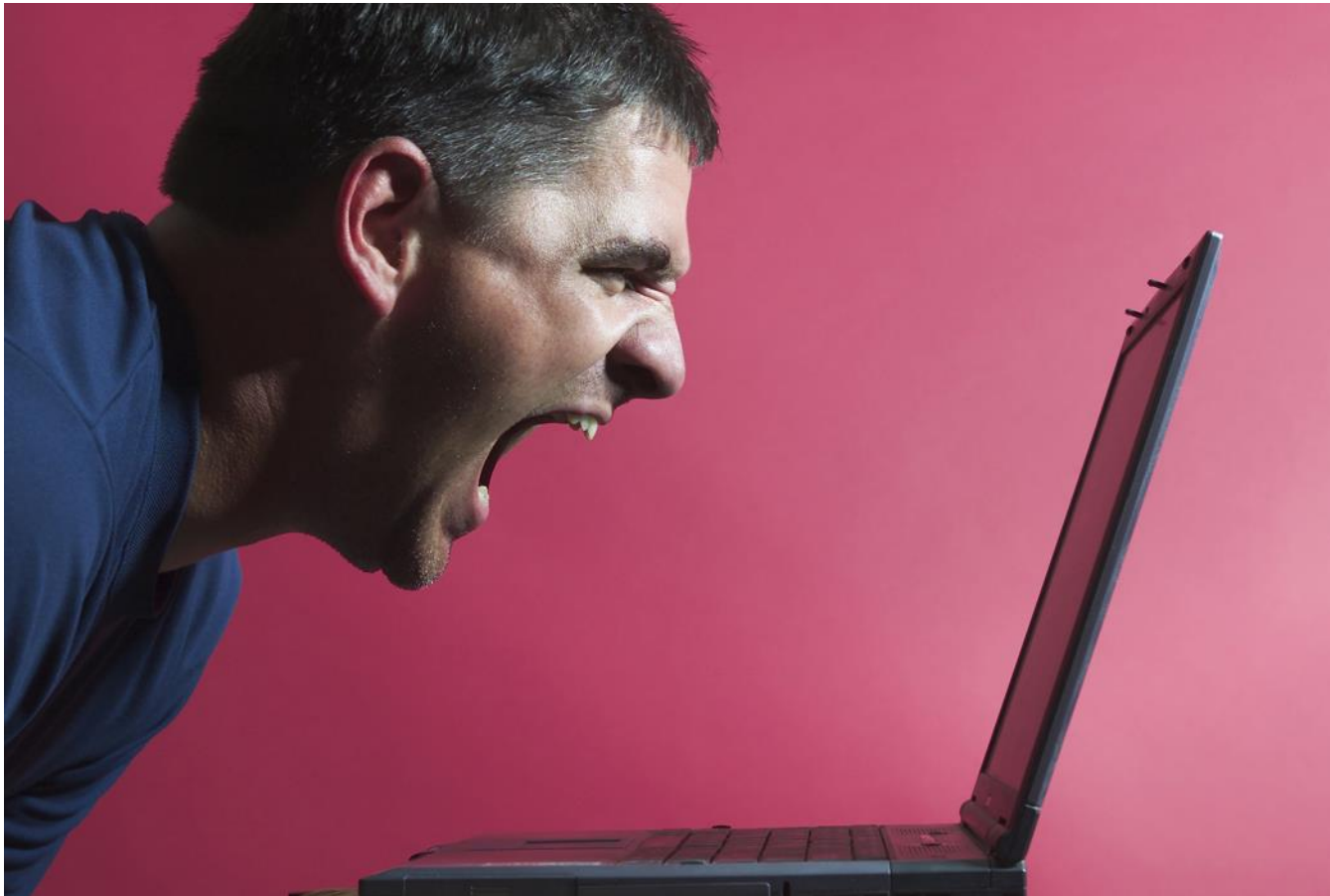
- **Information**

- Our email group: (I will add everyone's email to this group later)
- Github team: this is where we share materials and also have all important links for our program. Please bookmark it.

- **Setup:**

- We will use `ide.usaco.guide` or github codespace. Either way works.

# Why should we study computer science?



- A. Because computers are smart
- B. Because humans are smart
- C. Because computers are fast
- D. Because humans are fast
- E. Some combinations of the above

# How to calculate $1 + 2 + 3 + 4 + \dots + 100$ ?

- Human
- Computer

# Computer Classification



personal computer  
2 billion op/second



server  
about 10x faster than a normal PC



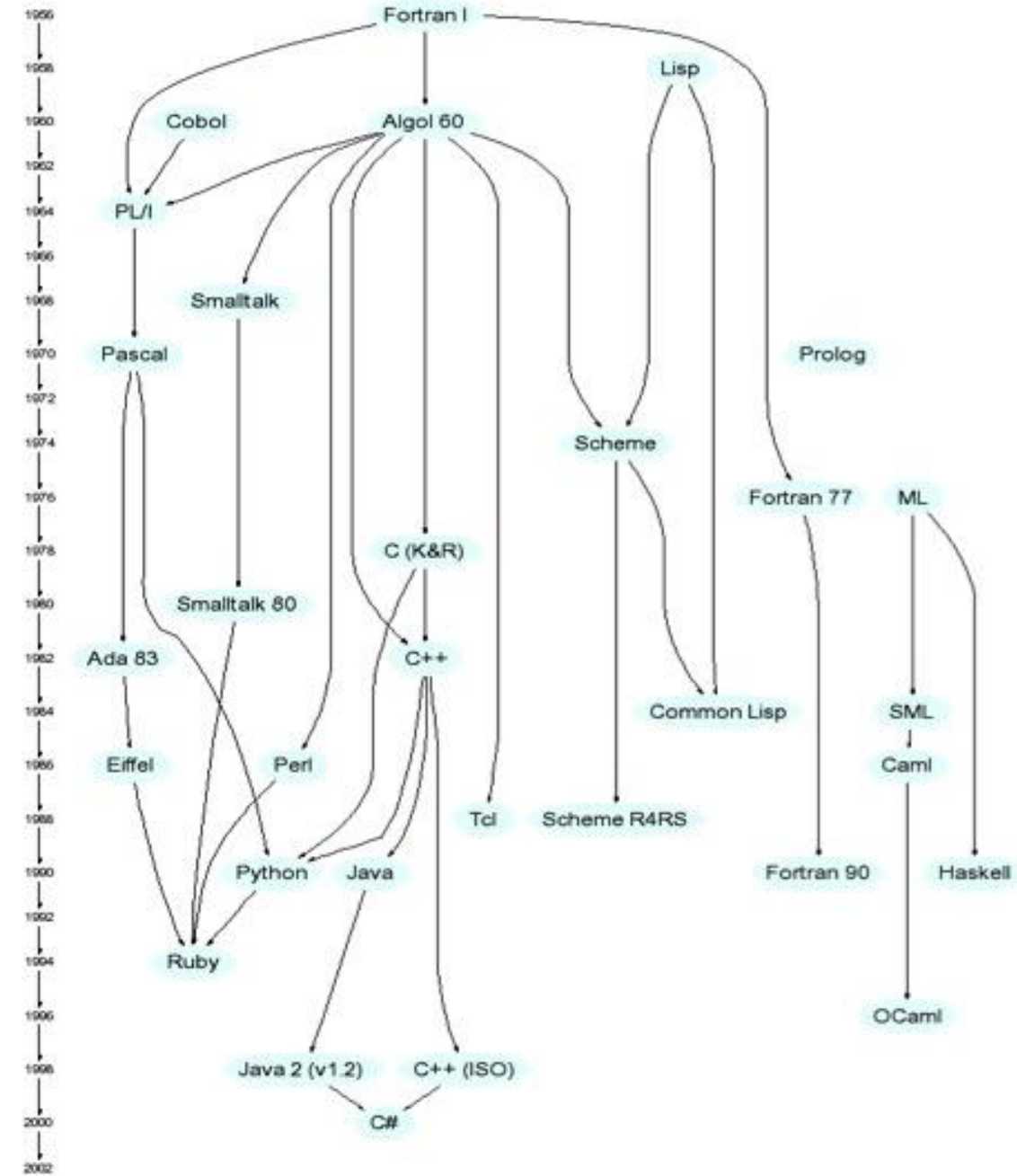
super computer  
about 146Mx faster than a normal PC

# Computer hardware

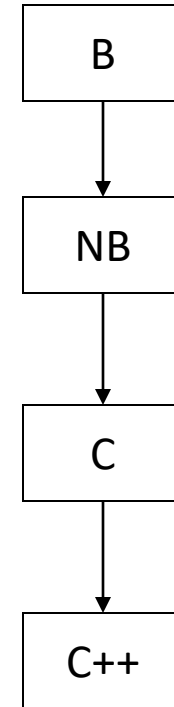
What are the major components of a computer?

- A. CPU
- B. Memory
- C. Disc
- D. Keyboard/monitor
- E. All of the above

# Programming Languages



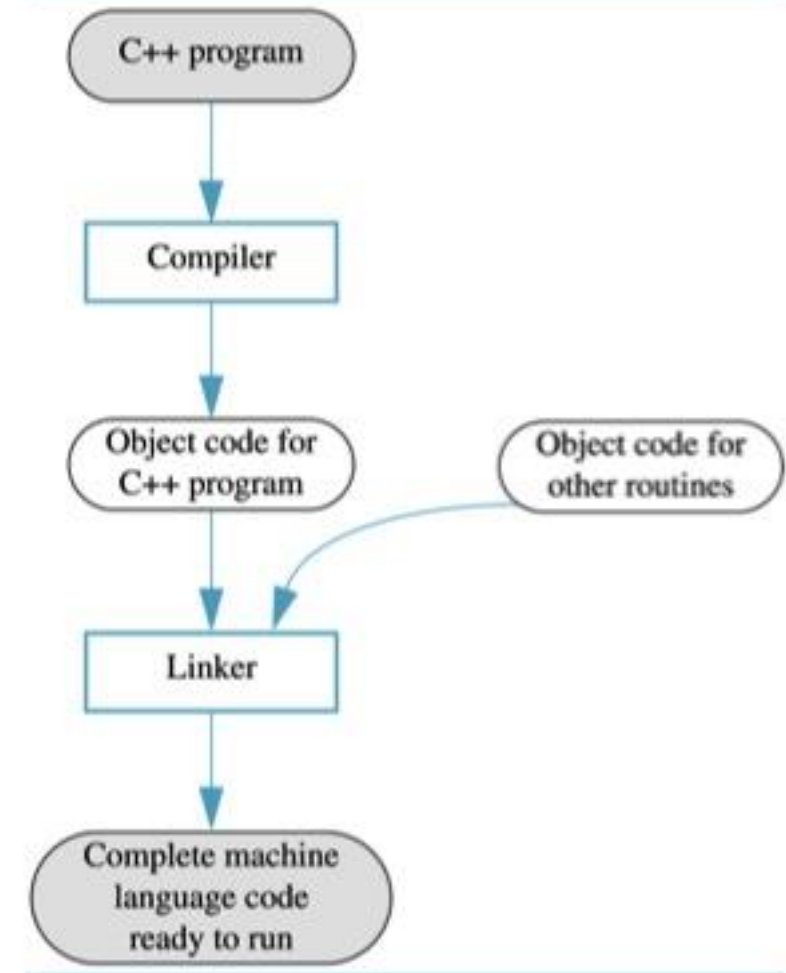
# Bjarne Stroustrup



# C++

```
#include <bits/stdc++.h>
using namespace std;
int main(){
    cout<<"hello world" <<endl;
}
```

## Preparing a C++ Program for Running



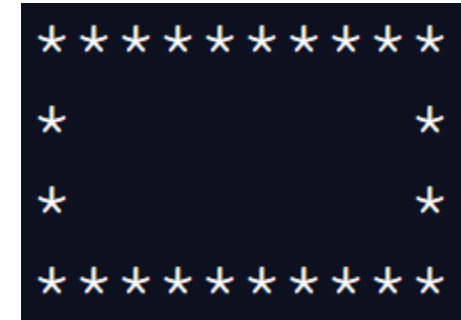
# What is wrong with the following code

I want to print out hello once

```
include <bits/stdc++.h>
using namespace std

main {
    cout<<hello<<endl
}
```

# Which code will print the rectangle to the right?



A

```
#include <iostream>
using namespace std;
int main() {
    cout << "*****";
    cout << " *      *";
    cout << " *      *";
    cout << "*****";
}
```

B

```
#include <iostream>

int main() {
    cout << "*****"<<endl;
    cout << " *      *"<<endl;
    cout << " *      *"<<endl;
    cout << " *      *"<<endl;
    cout << "*****"<<endl;
}
```

C

```
#include <iostream>
using namespace std;
int main() {
    cout << "*****"<<endl;
    cout << " *      *"<<endl;
    cout << " *      *"<<endl;
    cout << " *      *"<<endl;
    cout << "*****"<<endl;
}
```

D

```
using namespace std;
int main() {
    cout << "*****"<<endl;
    cout << " *      *"<<endl;
    cout << " *      *"<<endl;
    cout << " *      *"<<endl;
    cout << "*****"<<endl;
}
```

E. More than one of them will work

# What will the following code print out?

```
#include <iostream>
using namespace std;
int main() {
    cout<<"hell";
    cout<<"o"<<endl;
    cout<<"worl"<<endl;
    cout<<"d";
}
```

- A. hello world
- B. hello  
world
- C. hello  
worl  
d
- D. none of the above

# Exercise/Homework

- See the separate pdf file